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Course Code: CSA0903

Course Name: Programming in JAVA

Pattern Programs

1. Right-Angled Triangle of Stars

Java Program:

```
import java.util.Scanner;

public class RightAngledTriangle {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        for (int i = 1; i <= rows; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Sample Output:

```
*
**
***
****
*****
```

2. Number Pyramid

Java Program:

```
import java.util.Scanner;

public class NumberPyramid {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        for (int i = 1; i <= rows; i++) {
            for (int space = 1; space <= rows - i; space++) {
                System.out.print(" ");
            }
            for (int j = 1; j <= i; j++) {
```

```

        System.out.print(j + " ");
    }
    System.out.println();
}
}
}

```

Sample Output:

```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

3. Hollow Square Pattern

Java Program:

```

public class HollowSquare {
    public static void main(String[] args) {
        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= 5; j++) {
                if (i == 1 || i == 5 || j == 1 || j == 5)
                    System.out.print("*");
                else
                    System.out.print(" ");
            }
            System.out.println();
        }
    }
}

```

Sample Output:

```

*****
*   *
*   *
*   *
*   *
*****

```

4. Number Right-Angled Triangle

Java Program:

```

public class NumberRightAngledTriangle {
    public static void main(String[] args) {
        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(j + " ");
            }
            System.out.println();
        }
    }
}

```

Sample Output:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

5. Alphabet Triangle Pattern

Java Program:

```
public class AlphabetTriangle {
    public static void main(String[] args) {
        for (int i = 1; i <= 5; i++) {
            char ch = 'A';
            for (int j = 1; j <= i; j++) {
                System.out.print(ch + " ");
                ch++;
            }
            System.out.println();
        }
    }
}
```

Sample Output:

```
A
A B
A B C
A B C D
A B C D E
```

6. Diamond Star Pattern

Java Program:

```
public class DiamondStar {
    public static void main(String[] args) {
        int n = 3;

        for (int i = 1; i <= n; i++) {
            for (int space = 1; space <= n - i; space++)
                System.out.print(" ");
            for (int j = 1; j <= 2 * i - 1; j++)
                System.out.print("*");
            System.out.println();
        }

        for (int i = n - 1; i >= 1; i--) {
            for (int space = 1; space <= n - i; space++)
```

```

        System.out.print(" ");
        for (int j = 1; j <= 2 * i - 1; j++)
            System.out.print("*");
        System.out.println();
    }
}

```

Sample Output:

```

  *
 ***
*****
 ***
  *

```

7. Floyd's Triangle

Sample Output:

```

1
2 3
4 5 6
7 8 9 10
11 12 13 14 15

```

Java Program:

```

public class FloydsTriangle {
    public static void main(String[] args) {
        int num = 1;
        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(num + " ");
                num++;
            }
            System.out.println();
        }
    }
}

```

8. Right-Aligned Star Triangle

Java Program:

```

public class RightAlignedStarTriangle {
    public static void main(String[] args) {
        for (int i = 1; i <= 5; i++) {
            for (int space = 1; space <= 5 - i; space++)
                System.out.print(" ");
            for (int j = 1; j <= i; j++)
                System.out.print("*");
            System.out.println();
        }
    }
}

```

Sample Output:

```
*
**
***
****
*****
```

9. Inverted Number Triangle

Java Program:

```
public class InvertedNumberTriangle {
    public static void main(String[] args) {
        for (int i = 5; i >= 1; i--) {
            for (int j = 1; j <= i; j++) {
                System.out.print(j + " ");
            }
            System.out.println();
        }
    }
}
```

Sample Output:

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

10. Binary Number Triangle

Java Program:

```
public class BinaryNumberTriangle {
    public static void main(String[] args) {
        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print((i + j) % 2 + " ");
            }
            System.out.println();
        }
    }
}
```

Sample Output:

```
1
0 1
1 0 1
0 1 0 1
1 0 1 0 1
```

11. Half Diamond Star Pattern

Java Program:

```
public class HalfDiamondStar {
    public static void main(String[] args) {
        for (int i = 1; i <= 3; i++) {
            for (int j = 1; j <= i; j++)
                System.out.print("*");
            System.out.println();
        }

        for (int i = 2; i >= 1; i--) {
            for (int j = 1; j <= i; j++)
                System.out.print("*");
            System.out.println();
        }
    }
}
```

Sample Output:

```
*
**
***
**
*
```

12. Hollow Right-Angled Triangle

Java Program:

```
public class HollowRightAngledTriangle {
    public static void main(String[] args) {
        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= i; j++) {
                if (j == 1 || j == i || i == 5)
                    System.out.print("*");
                else
                    System.out.print(" ");
            }
            System.out.println();
        }
    }
}
```

Sample Output:

```
*
**
* *
* *
*****
```

13. Reverse Pyramid of Stars

Java Program:

```
public class ReversePyramid {
```

```

    public static void main(String[] args) {
        int rows = 5;
        for (int i = rows; i >= 1; i--) {
            for (int space = 0; space < rows - i; space++)
                System.out.print(" ");
            for (int j = 1; j <= 2 * i - 1; j++)
                System.out.print("*");
            System.out.println();
        }
    }
}

```

Sample Output:

```

*****
*****
****
***
**
*

```

14. Continuous Characters Triangle

Java Program:

```

public class ContinuousCharactersTriangle {
    public static void main(String[] args) {
        char ch = 'A';
        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(ch + " ");
                ch++;
            }
            System.out.println();
        }
    }
}

```

Sample Output:

```

A
B C
D E F
G H I J
K L M N O

```

15. Right-Angled Triangle with Even Numbers

Java Program:

```

public class EvenNumberTriangle {
    public static void main(String[] args) {
        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(2 * j + " ");
            }
            System.out.println();
        }
    }
}

```

```
}  
}
```

Sample Output:

```
2  
2 4  
2 4 6  
2 4 6 8  
2 4 6 8 10
```

16. Triangle of Odd Numbers

Java Program:

```
public class OddNumberTriangle {  
    public static void main(String[] args) {  
        int num = 1;  
        for (int i = 1; i <= 5; i++) {  
            for (int j = 1; j <= i; j++) {  
                System.out.print(num + " ");  
                num += 2;  
            }  
            System.out.println();  
        }  
    }  
}
```

Sample Output:

```
1  
3 5  
7 9 11  
13 15 17 19  
21 23 25 27 29
```

17. Hollow Pyramid Pattern

Java Program:

```
public class HollowPyramid {  
    public static void main(String[] args) {  
        int rows = 5;  
        for (int i = 1; i <= rows; i++) {  
            for (int space = 1; space <= rows - i; space++)  
                System.out.print(" ");  
            for (int j = 1; j <= 2 * i - 1; j++) {  
                if (j == 1 || j == 2 * i - 1 || i == rows)  
                    System.out.print("*");  
                else  
                    System.out.print(" ");  
            }  
            System.out.println();  
        }  
    }  
}
```



```
        }  
        System.out.println();  
    }  
}
```

Sample Output:

```
  *  
 * *  
*   *  
*     *  
*****
```